# Image Data Analysis & Machine Learning in *Orange*

Find our materials here→
https://tinyurl.com/CUHKImageworkshop

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## How many of you can do programming?

**How many of you enjoy programming?**

# What is *Orange*?

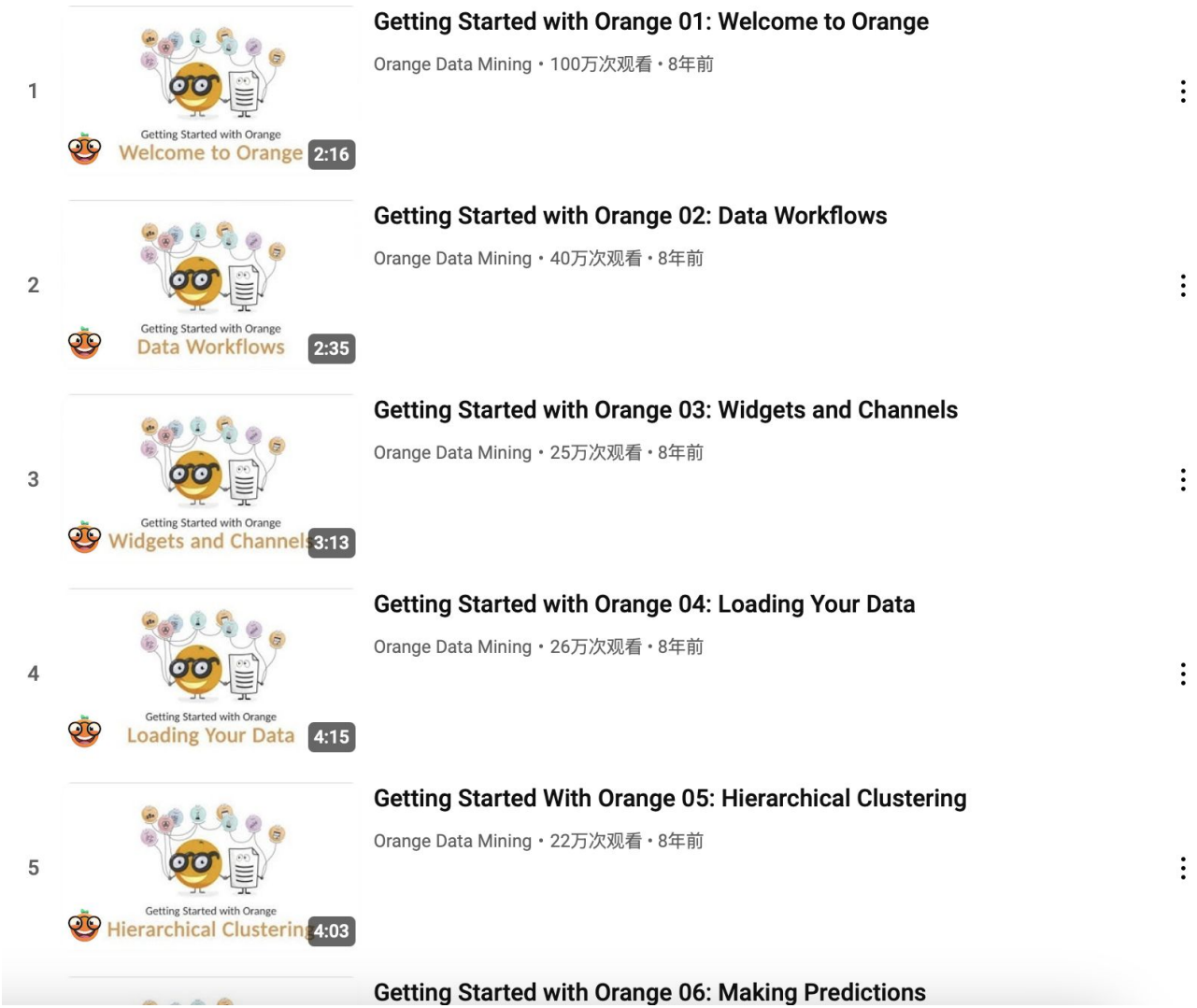
- an open-source data visualization and analysis tool.
- particularly for data mining, machine learning, and data visualization.
- ***Orange*** **makes everything simple!  :)**

# Video tutorials:

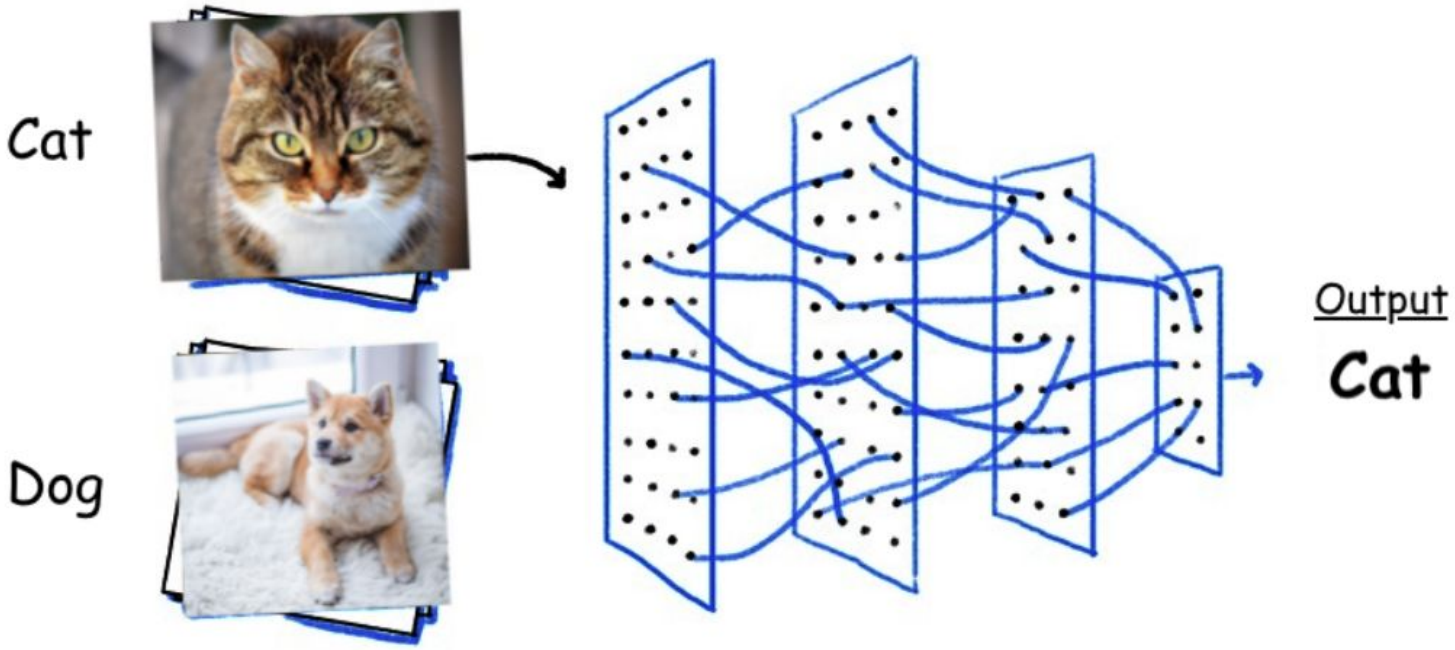
https://www.youtube.com/watch?v=lb-x36xqJ-E&list=PLmNPvQr9Tf-ZSDLwOzxpvY-HrE0yv-8Fy&index=2

You can start ***Orange*** from here!

The official website offers very detailed tutorials on creating the data visualizations you want step by step.



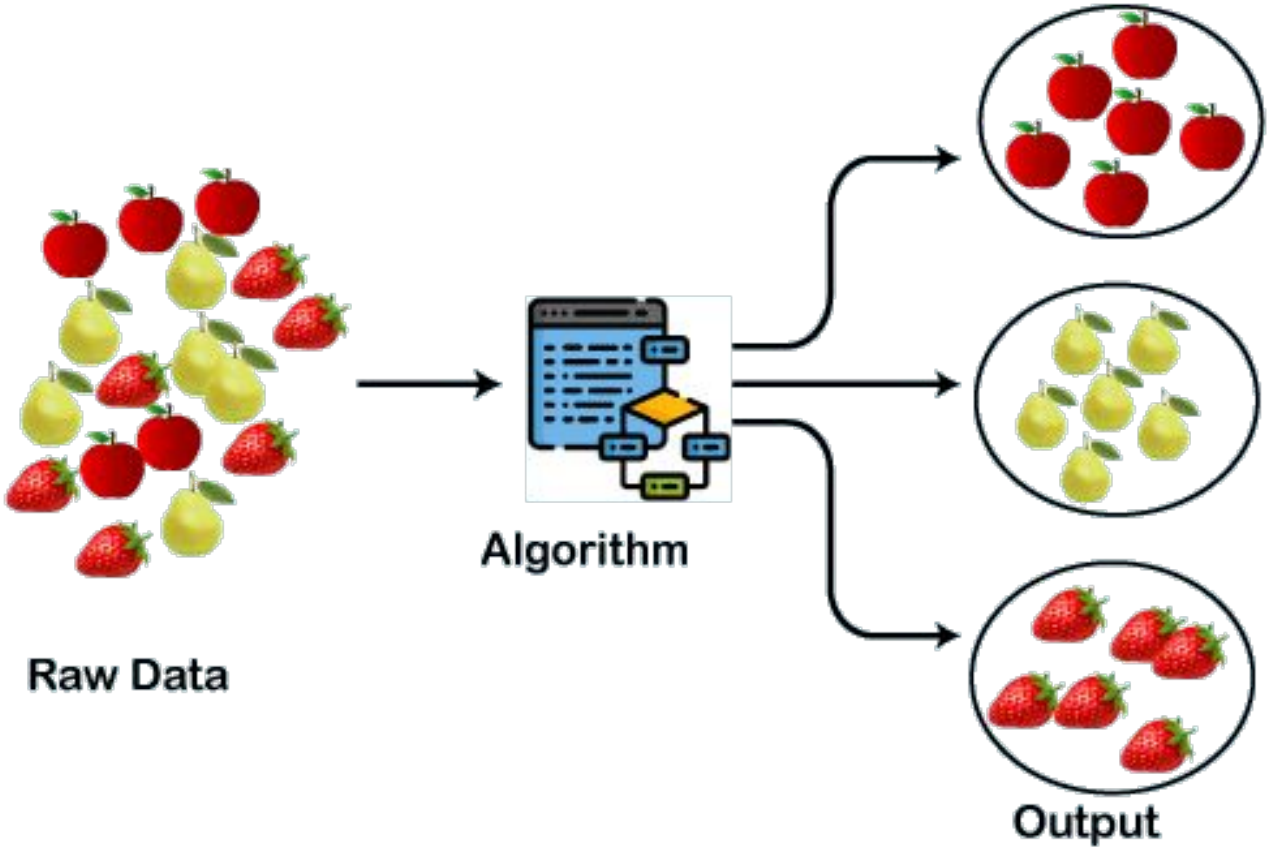
# Image Data Analysis



# What can Orange do with images?

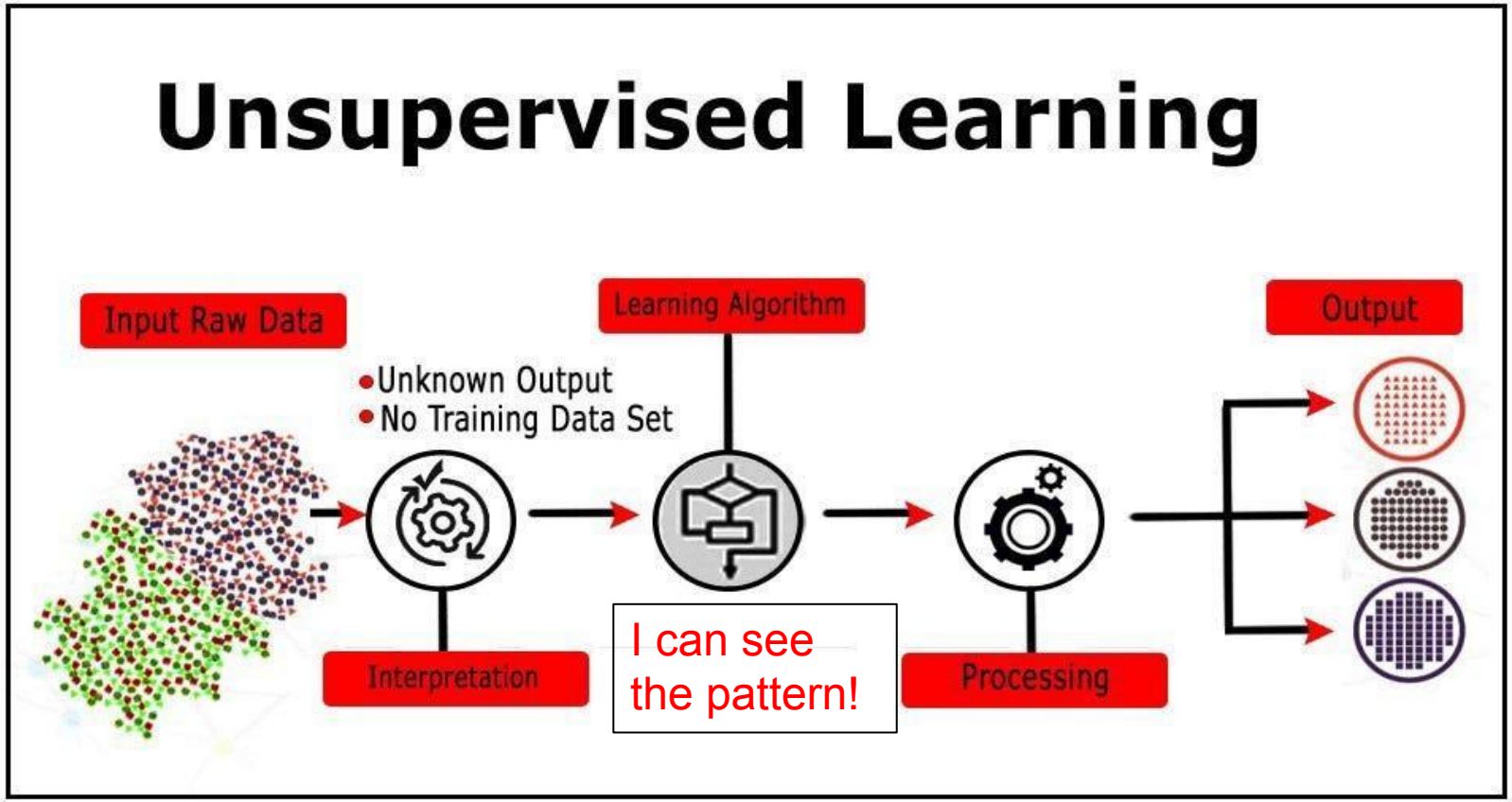
## Hierarchical Clustering

- An **unsupervised machine learning** technique used to group similar data points together.
- To find natural, hidden patterns or structures in data without any pre-existing labels.



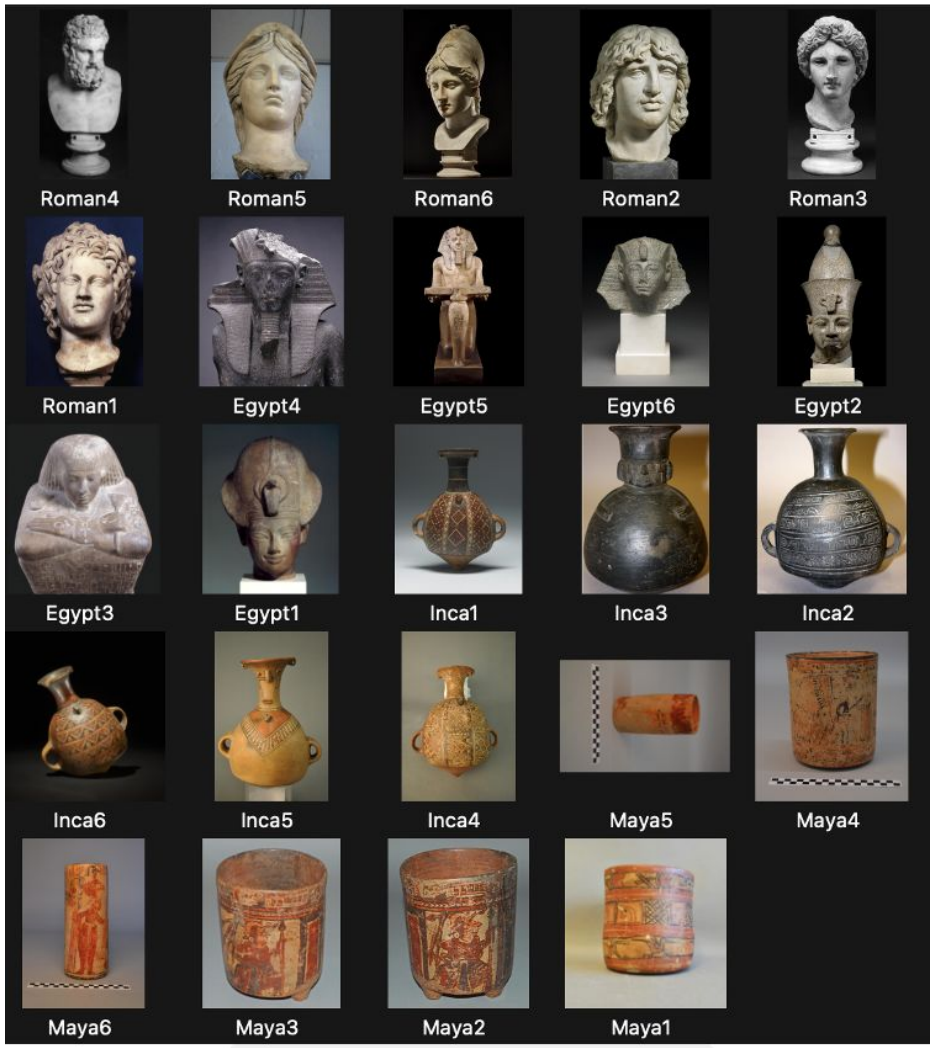
# Why is it “machine learning”?

- It's a data-driven approach; the machine “learns” the grouping rules from the data itself.
- The algorithm improves its grouping with each iteration to find the optimal structure.



# Today’s Goal

## Using hierarchical clustering to perform image analysis by Orange




Install - operate - interpret – self-practice – brainstorming

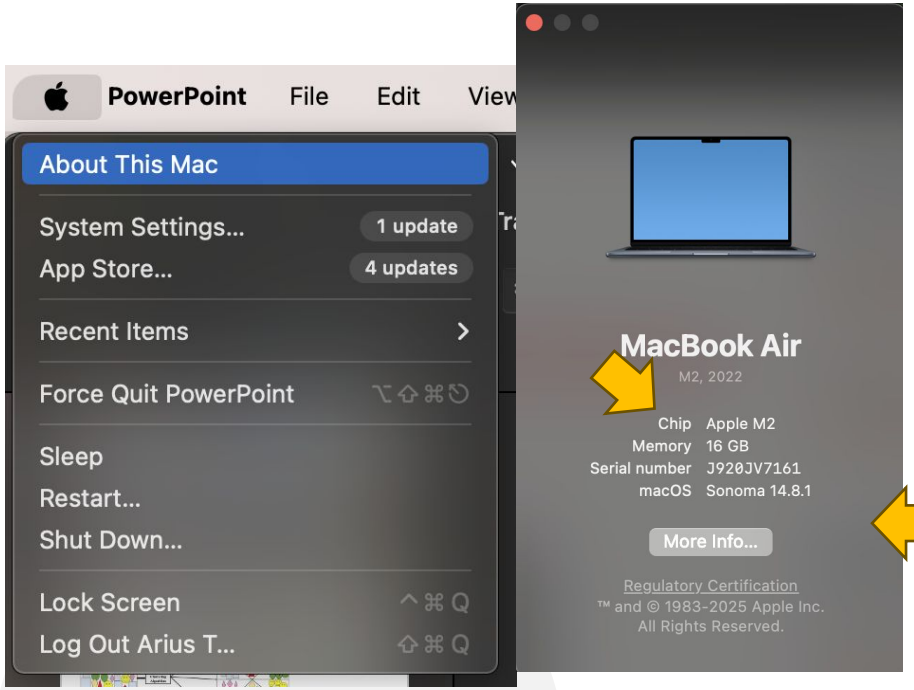
# Steps: 1. Install Orange

**Windows users: https://download.biolab.si/download/files/**

Orange3-3.38.0-Python3.10.11-x86_64.dmg
Orange3-3.38.0-Python3.11.8-arm64.dmg
Orange3-3.38.0-Python311-win_amd64.exe
Orange3-3.38.0.zip
Orange3-3.38.1-Miniforge-x86_64.exe
Orange3-3.38.1-Python3.10.11-x86_64.dmg
Orange3-3.38.1-Python3.11.8-arm64.dmg
Orange3-3.38.1-Python311-win_amd64.exe
Orange3-3.38.1-problematic-link.zip
Orange3-3.38.1.zip
Orange3-3.39.0-Python3.10.11-x86_64.dmg

**Download Orange3-3.38.1-Python311-win_amd64.exe**

**Apple users: https://orangedatamining.com/download/**




## macOS

### Orange for Apple silicon

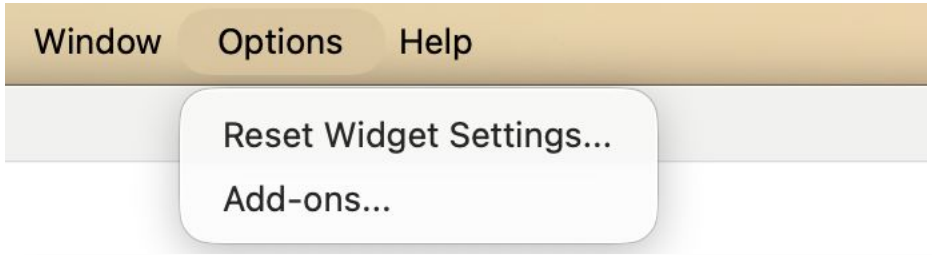
Orange3-3.40.0-Python3.12.10-arm64.dmg

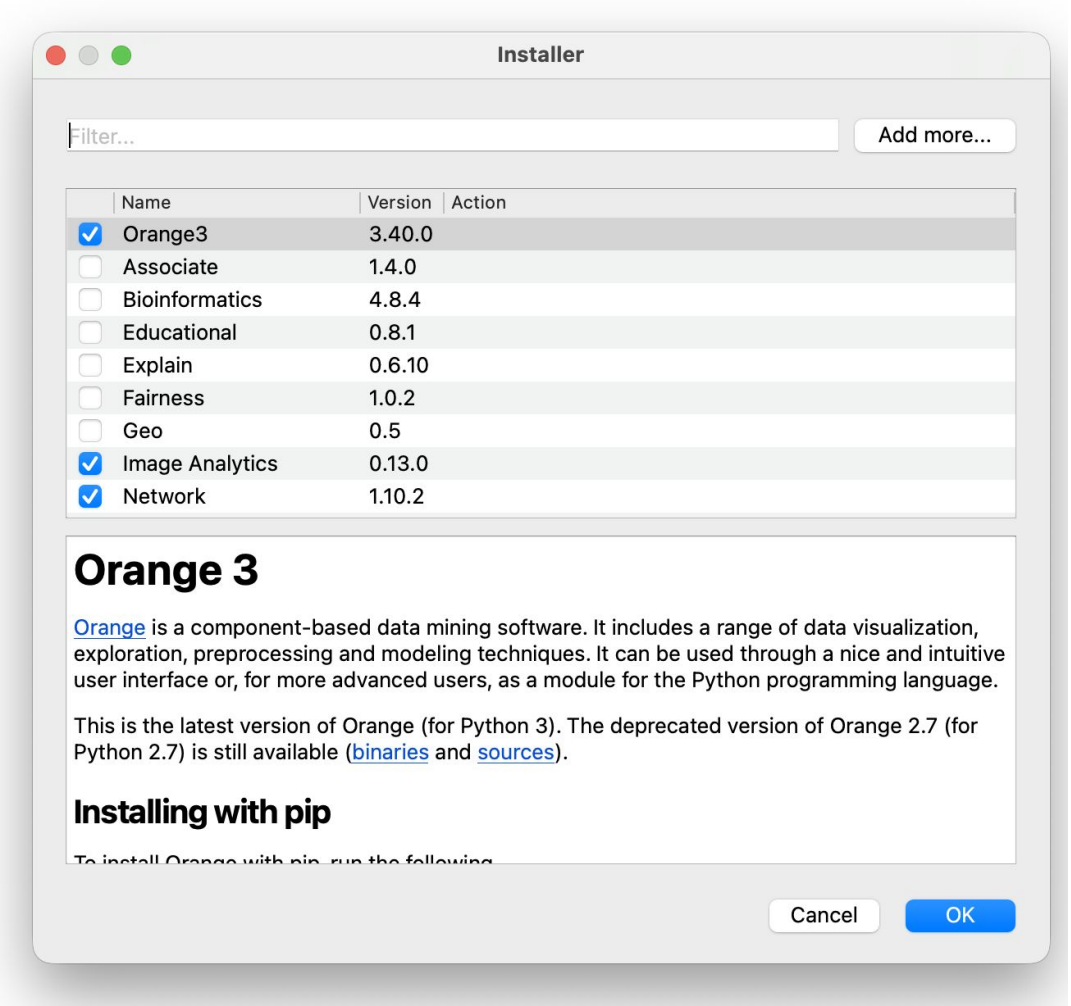
### Orange for Intel

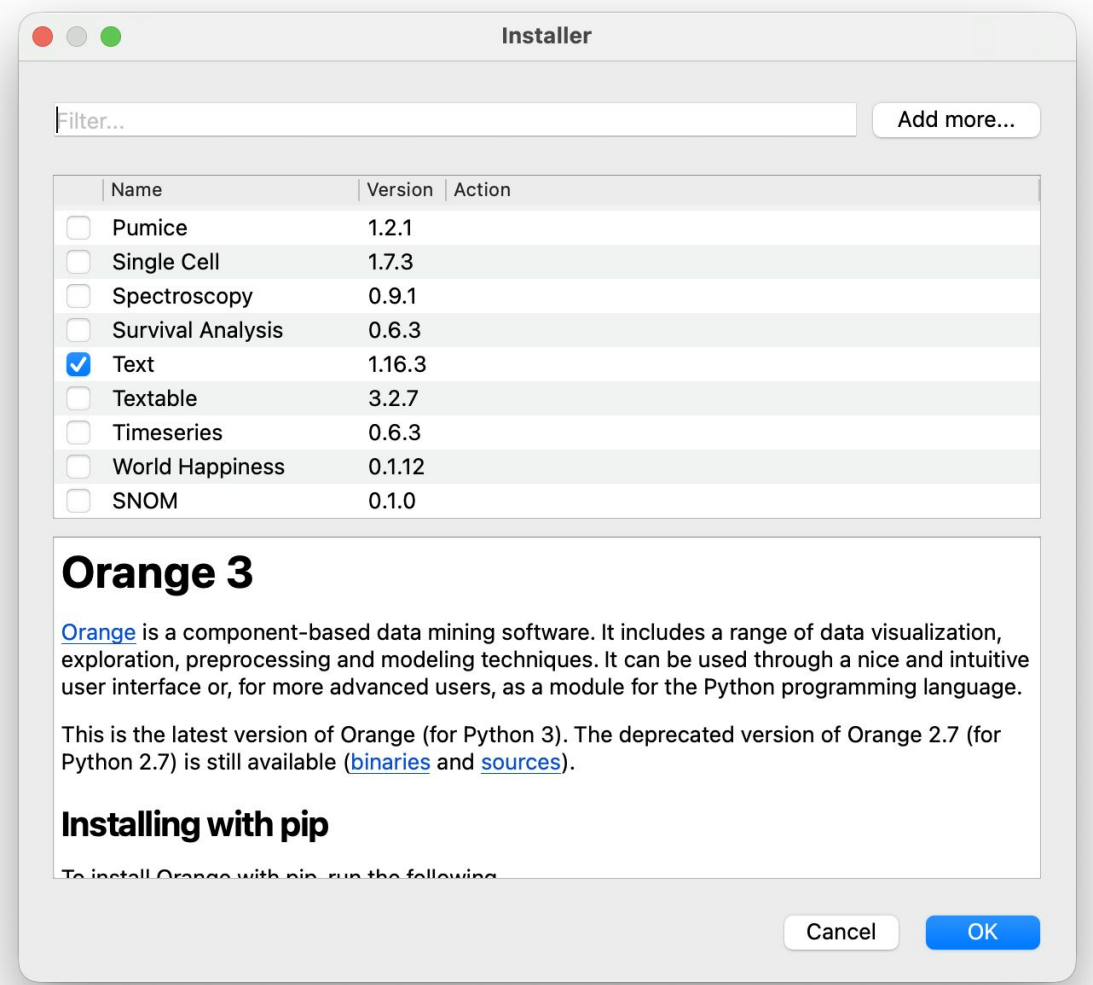
Orange3-3.40.0-Python3.12.10-x86_64.dmg

**Not sure which installer to select?** Click the Apple logo in the top-left corner of your screen, select About This Mac, and check the Chip or Processor field. If you see Apple, select the Orange for Apple Silicon installer. If you see Intel, select the Orange for Intel.

# 2. Open Orange- options - ‘add-ons’



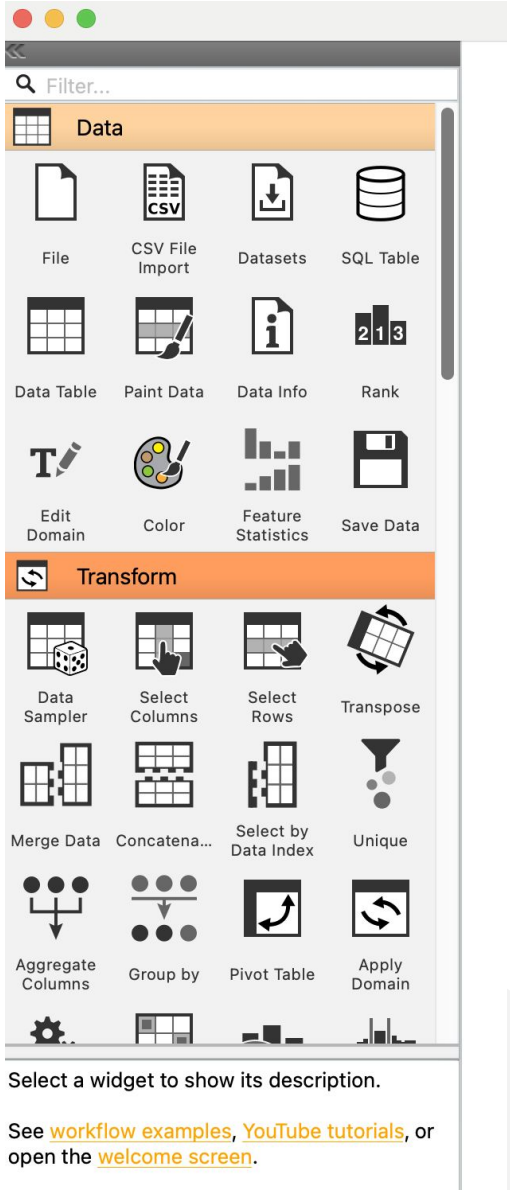
# 3. Download two image folders here:

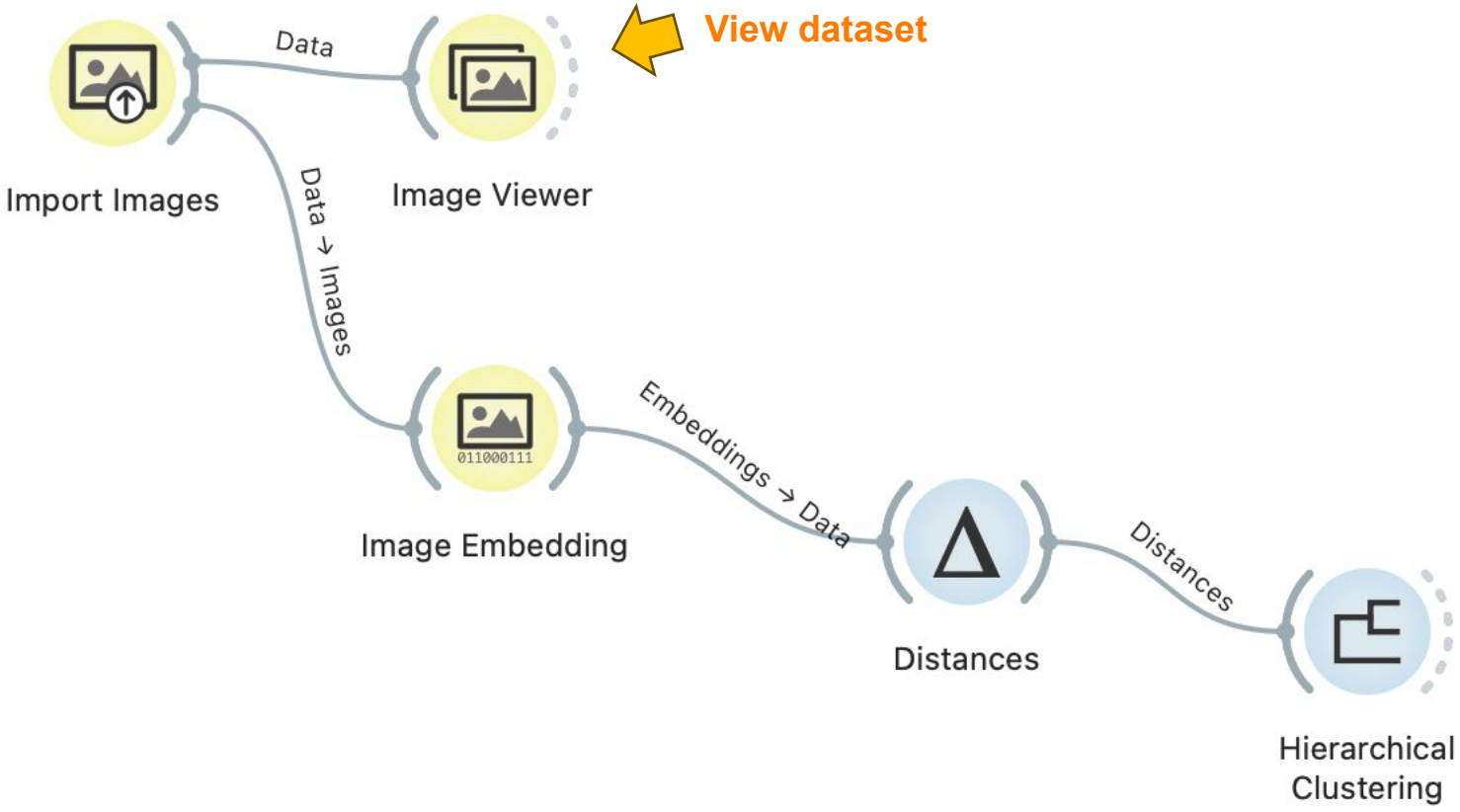
https://tinyurl.com/35z76rse

Islamic Coins VS Chinese Coins

Statues VS Vases

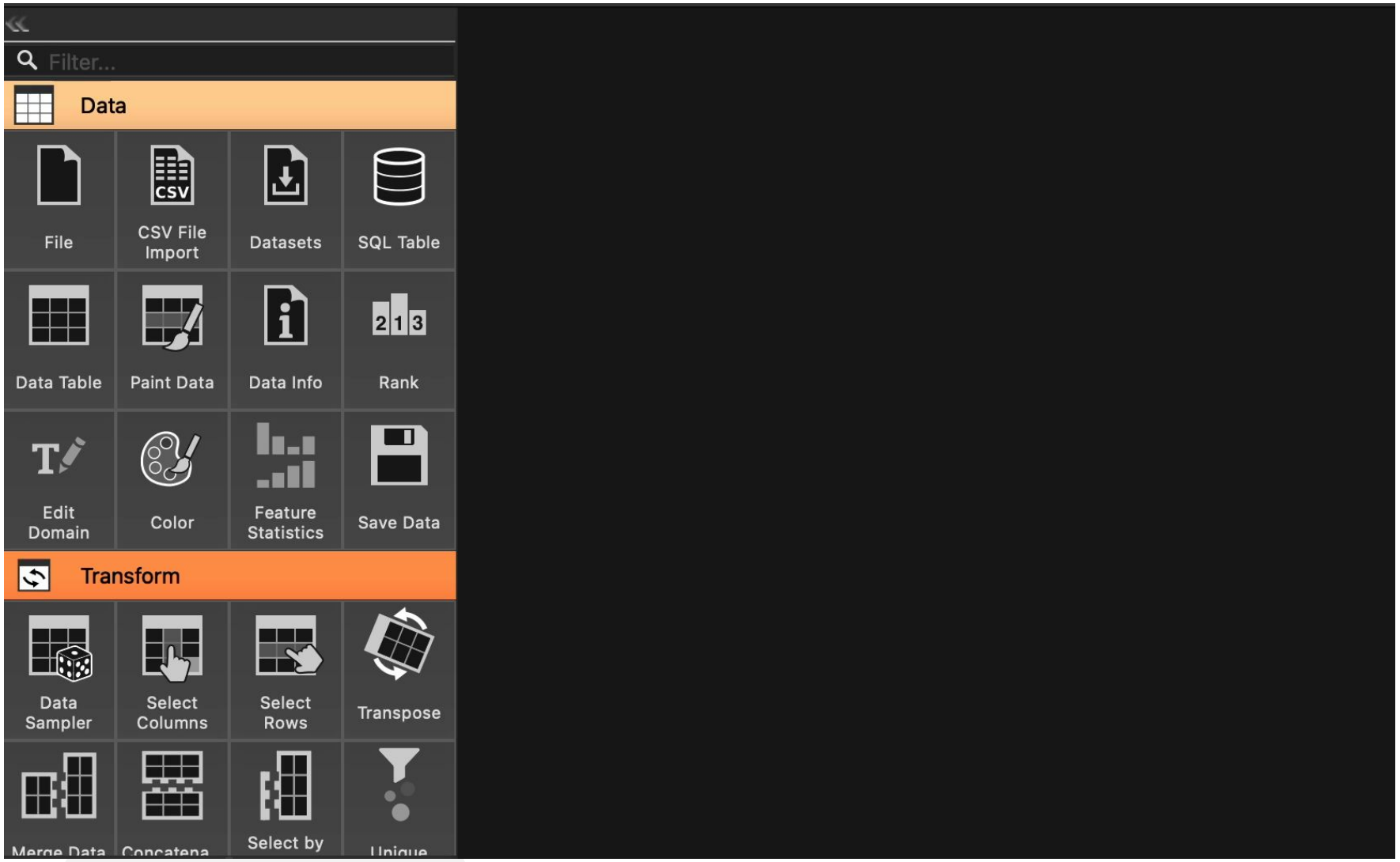
# 4. Use the folder “Islamic Coins VS Chinese Coins” Follow the procedure map below:



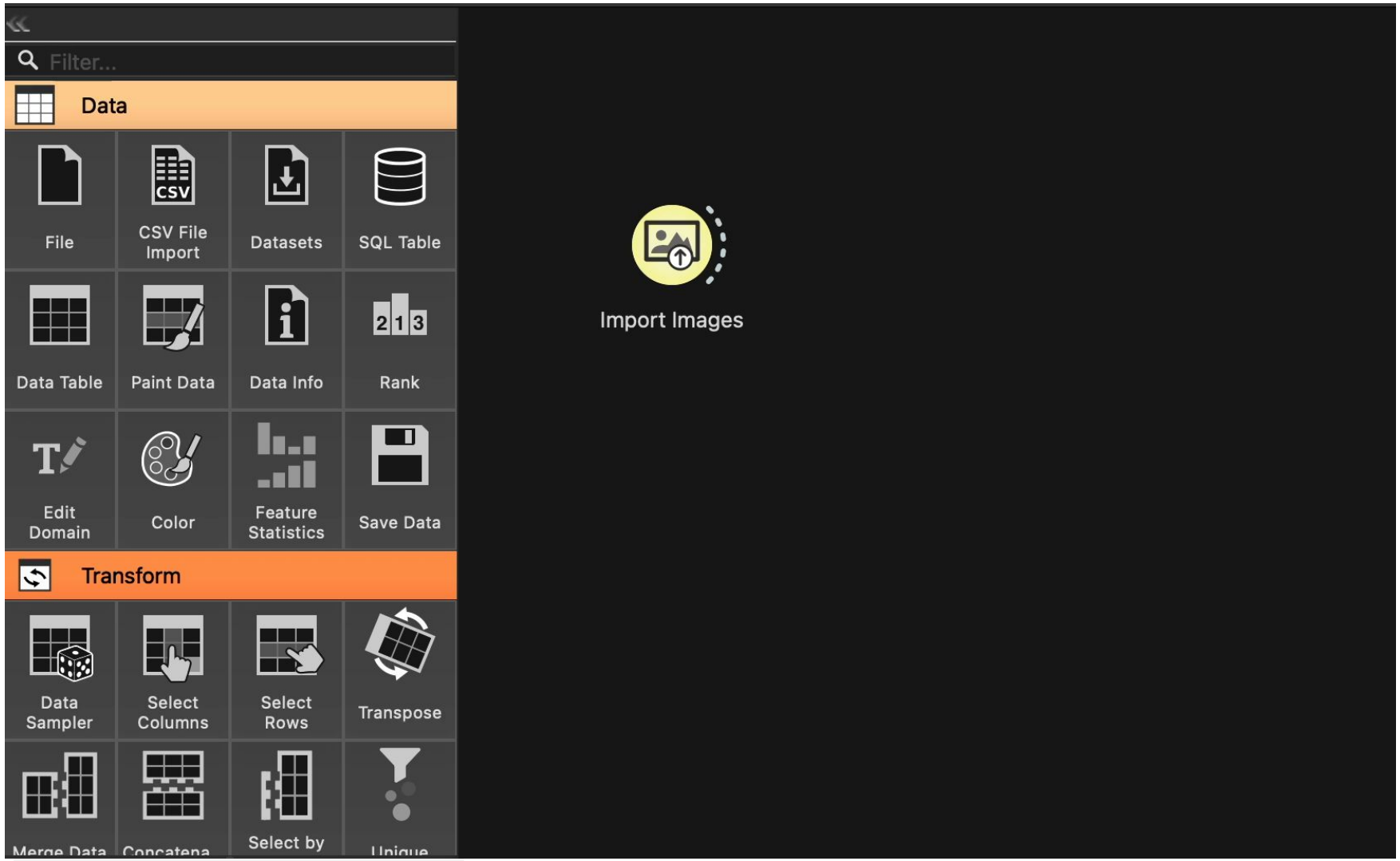
| Step | Purpose | Example (Islamic & Chinese Coins) |
|---|---|---|
| **Import images** | Establish the **raw dataset** of research objects | Import a mixed collection of Islamic and Chinese coin images to build the corpus for analysis |
| **Image embedding** | Transform **unstructured visual content** into **structured numerical features** (comparable "feature vectors") | Let the computer automatically extract visual characteristics of each coin (e.g., shape, script patterns, texture), generating comparable feature indicators |
| **Distances** | Quantify **similarity/dissimilarity** between samples | Calculate visual distances between all pairs of coins; smaller values indicate higher visual similarity, forming a similarity matrix |
| **Hierarchical Clustering** | Discover **underlying structures and categorical patterns** in the data through unsupervised learning | Allow coins to automatically group based on similarity, generating a “family tree” that reveals potential classification systems and exceptional samples |

# To create initial module:



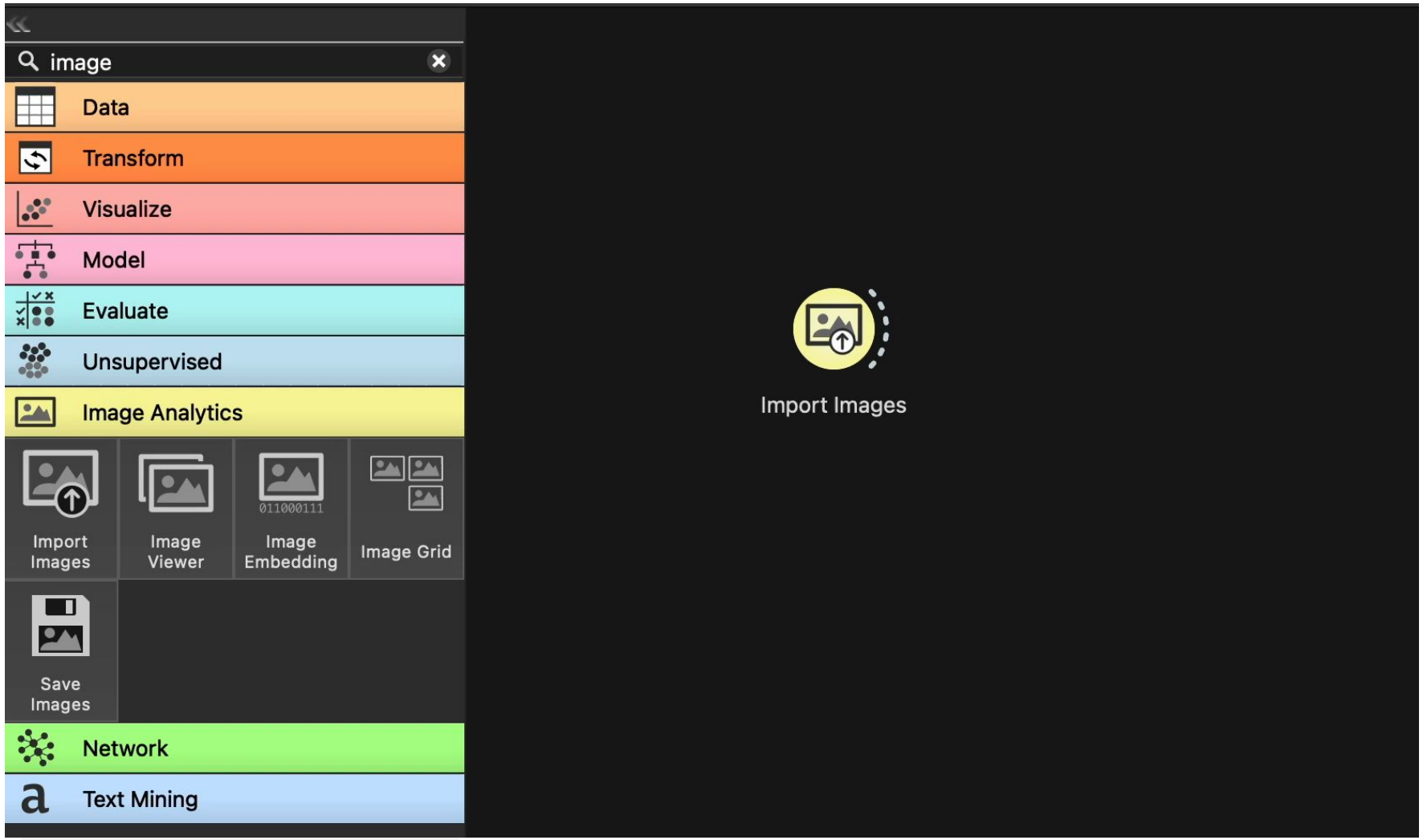
# To create next module(s):

## Method 1

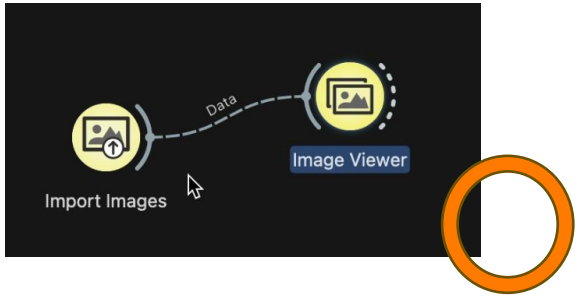


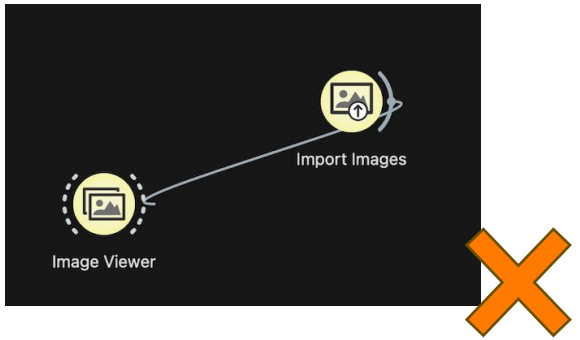
# To create next module(s):

## Method 2

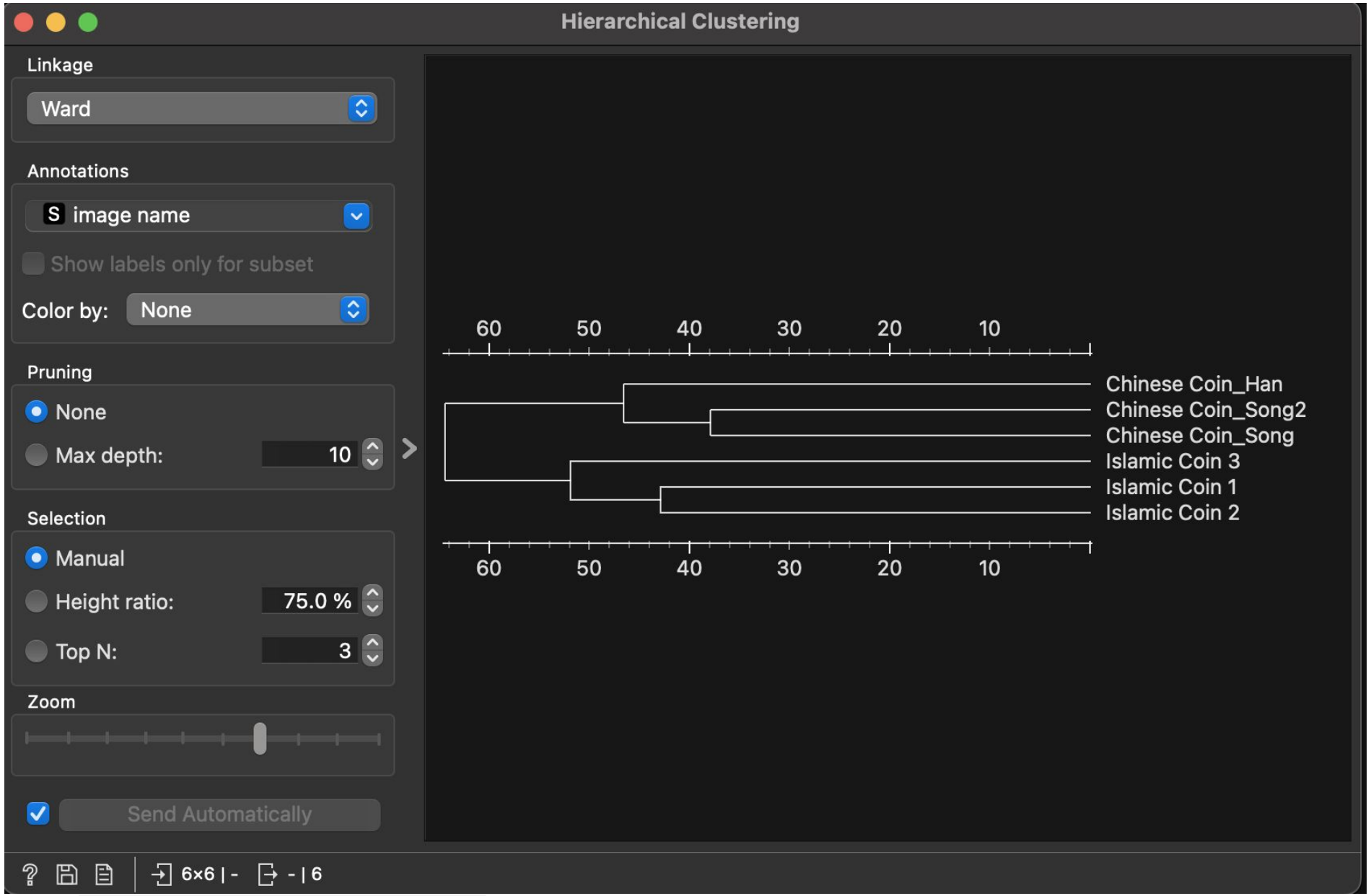



*Make sure the next module is on the right side and connected correctly!

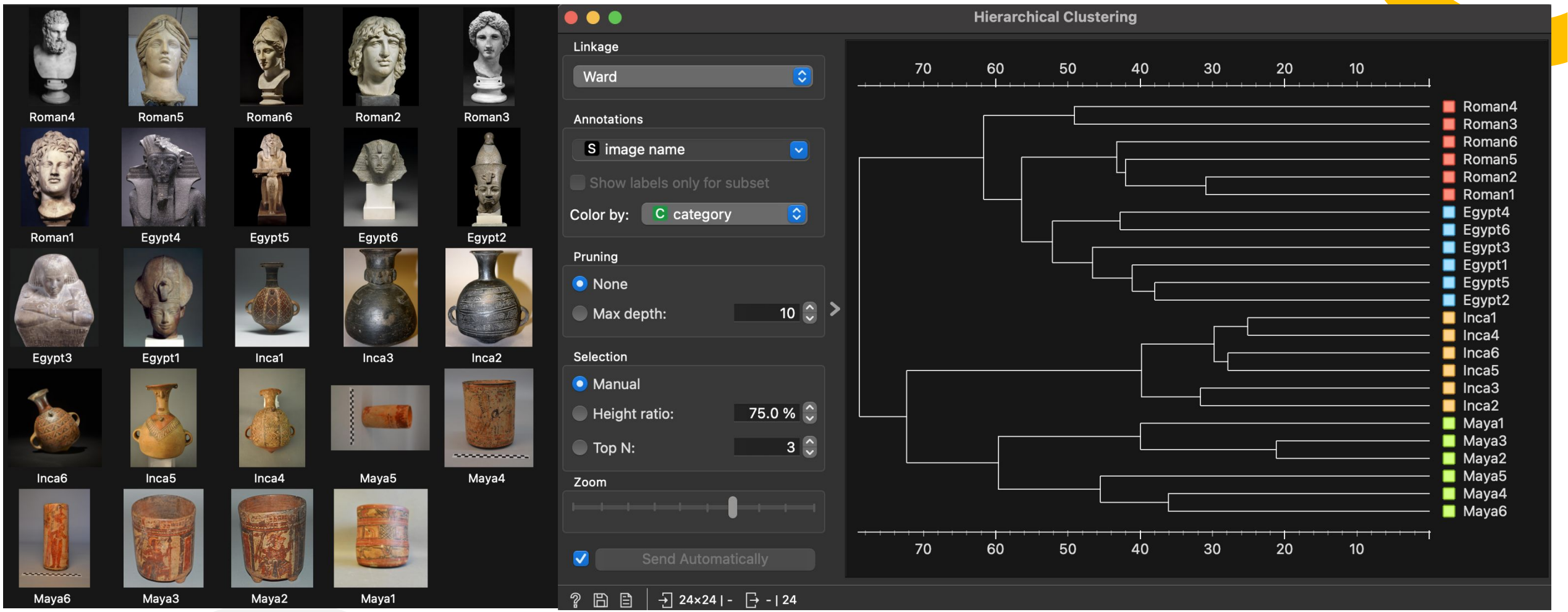


# 5. Double click “Hierarchical Clustering”



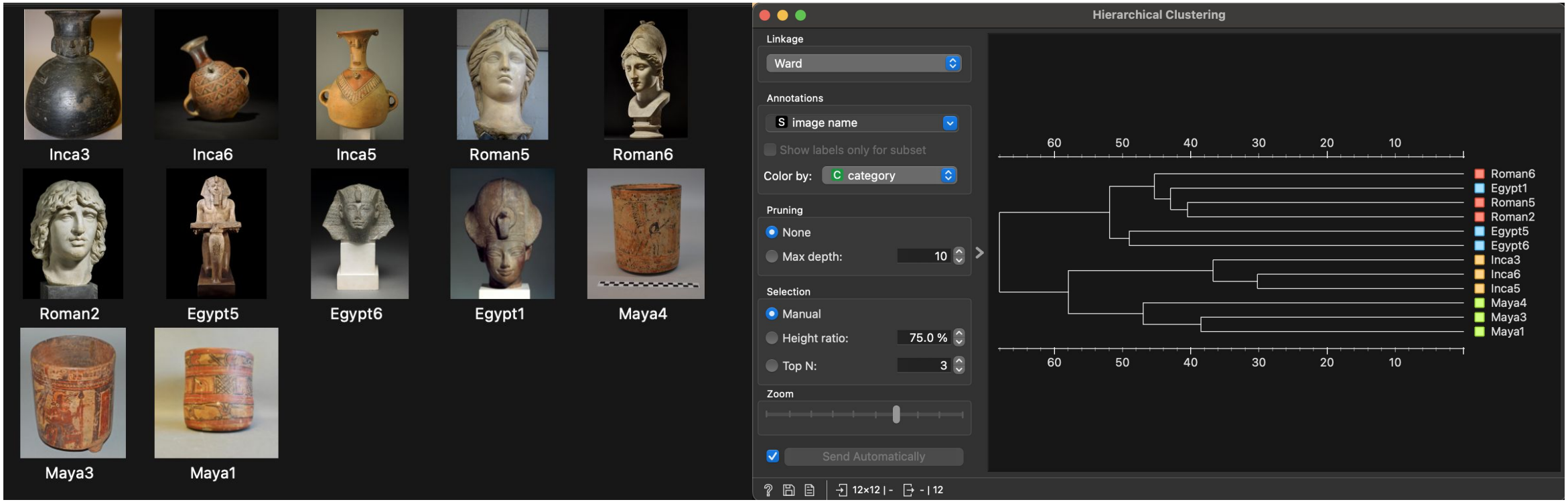
# 6. Use the folder ”Statues VS Vases” to self-practice




**Questions:**

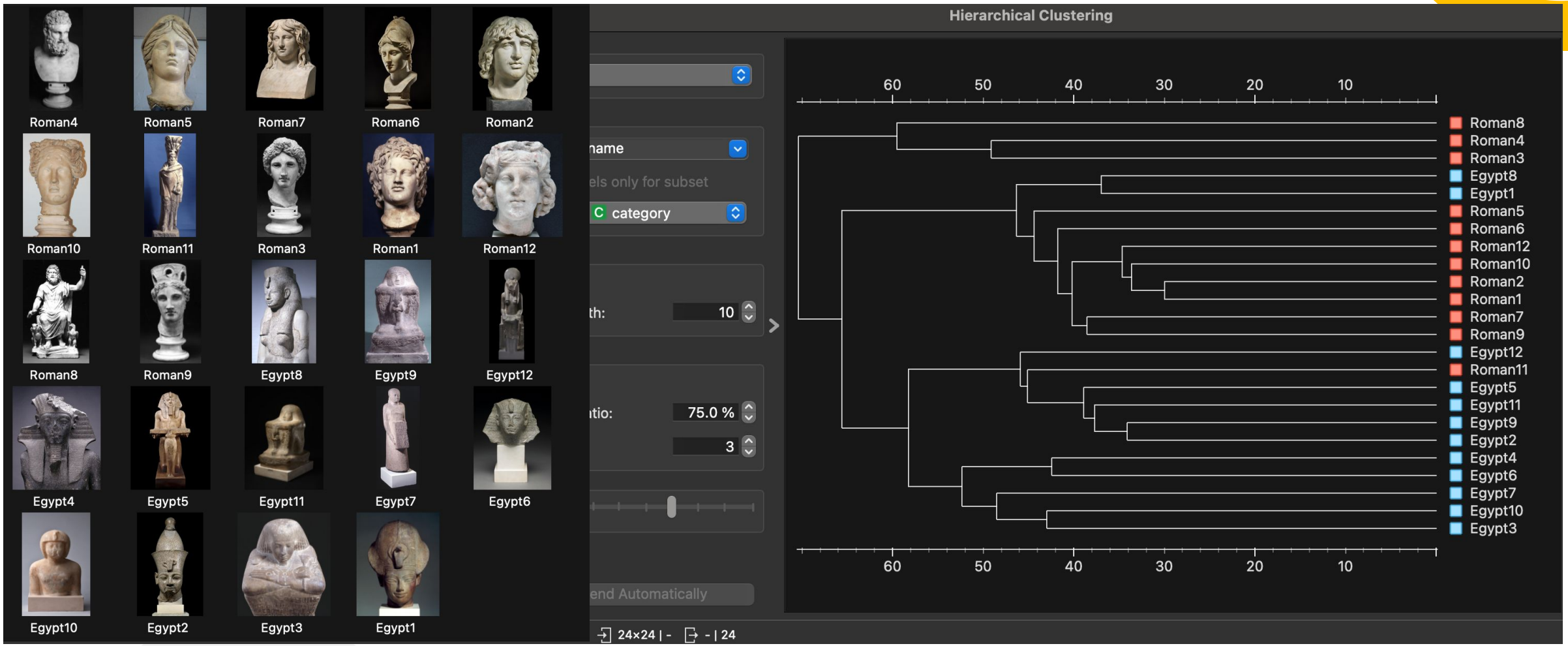
1. How can we replace existing dataset (Coins) with the new dataset?
2. How can we “create” the colour blocks on the clustering network?

# Advanced Question- case 1:




**Why is the result inaccurate (i.e. Egypt and Roman statues sorted into one category)?**

# Advanced Question- case 2:




**Why is the result inaccurate (i.e. Egypt and Roman statues sorted into one category)?**

There are even more exciting features waiting for you to explore!

**Get started now** and embark on your exploration journey!

# Thank you!

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